

JOHNSON MUGARRA

Data Scientist and Research Consultant

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PROFESSIONAL SUMMARY

Data scientist and research consultant with an M.Sc. in Data Science and six years working with field and operational data across East Africa. At Enimiro I design studies (child labor, living income, pollination) and build the pipelines, dashboards, and yield models behind them. At the Center for Research and Communication I coordinate proposals and lead the technical work on evaluations covering design, sampling, tools, data quality, analysis, and reporting. Earlier roles cover vanilla value-chain forecasting with Makerere University and MAAIF, process evaluation at IPA, and field research for USAID SITES.

CORE SKILLS

- Statistical software: Stata, R, Python, SQL, SAS, SPSS, Excel
- Research and evaluation: survey design, piloting and translation, computer-assisted interviewing, key informant interviews, focus group discussions, transcription, process and impact evaluation, monitoring and evaluation frameworks, indicator tracking
- Data collection and quality: questionnaires, tablet-based collection, enumerator training and supervision, validation, backup and documentation
- Visualization and reporting: Power BI, Tableau, dashboards, reports for donors and management
- Data systems: SurveyCTO API, PostgreSQL, Google Sheets, ETL pipelines, GeoPandas
- Machine learning and AI: scikit-learn, time-series forecasting, neural networks and CNNs in TensorFlow and PyTorch, and daily use of AI tools for cleaning data, writing code, and drafting, with the output checked by hand
- Proposal development: concept note, strategy, design, methods and work plans, budgeting, quality assurance

WORK EXPERIENCE

Research and Data Consultant, November 2025 - October 2026

Enimiro Products Uganda Ltd, Kayunga, Uganda

- Coordinated the data team on collection, cleaning, and organization of operational data across farmer activities, crop production, sales, and sustainability.
- Maintained a Python pipeline moving SurveyCTO submissions into PostgreSQL raw, clean, and analysis tables, with scheduled exports to Google Sheets for dashboards.

- Designed studies from proposal to analysis, including the child labor study, the living income study against Anker benchmarks, pollination and maturity modules, and the M&E measurement plan.
- Matched GPS to district boundaries and maintained the farmer master with merge and change logs for audits.
- Built dashboards and reports tracking performance indicators for management and the board.
- Ran statistical and trend checks to flag problems and turned findings into training manuals and field guidance for staff.
- Led the data and M&E workstream: built the M&E measurement plan and monitoring toolkit (KPI dashboards, tier trackers, review packs) used across 19 districts, and managed the field-extension restructuring from concept note to roll-out.
- Ran crop and supply-chain research, updated survey tools, trained staff on data use, and documented methods for donor and certification reporting.

Technical Lead Consultant, January 2025 - Present

Center for Research and Communication (CRC), Remote

- Reviewed client requirements and evaluation criteria to set the proposal approach and coordinated specialist inputs against deadlines.
- Wrote and edited technical sections, including methods and work plans, and worked with finance to align technical approaches with compliant budgets.
- Led the technical work for the IRC MERP endline evaluation, covering design and tools, analysis and synthesis, and integration of qualitative and quantitative findings against the results framework.
- Led quantitative inputs for IRC baseline and perception studies: sampling, instrument testing, data-quality checks, and statistical analysis.
- Served as data and M&E lead for UNICEF behavior change and awareness studies: tools, trackers, datasets, reach analysis, and report drafting.

Researcher / Data Scientist, Vanilla Value-Chain, January 2024 - June 2025

Makerere University / MAAIF, Catholic Relief Services, and sector partners, Uganda

- Researched vanilla maturity and harvest timing for national harvest-declaration considerations, combining field and crop work with statistical modeling and thermal-time analysis.
- Integrated field, satellite climate, and survey data and built time-series models to compare maturity-assessment methods and their timing, quality, and price implications.
- Conducted interviews with farmers and partners, field measurements, and seasonal maturity assessments across vanilla-growing districts.

- Established the Kabanyolo demonstration plot with the project team and prepared models, scenario notes, and technical inputs for supervisors and project partners. Work concluded with program funding closure.

Research Associate, February 2023 - November 2023

Innovations for Poverty Action (IPA), Kampala, Uganda

- Developed and implemented process evaluation plans with program managers.
- Designed survey instruments and supported piloting, translation, and computer-assisted interviewing.
- Coordinated evaluation logistics including timelines, field staffing, and procurement, and managed data entry, cleaning, and analysis.
- Applied data-quality protocols including enumerator audits and monitoring of data collection.
- Maintained relationships with partner organizations, contributed to meetings and reports, and supported research approvals and compliance.

Qualitative Research Associate, July 2023 - March 2024

USAID Strategic Information Technical Support (SITES), Uganda

- Trained on field methods and tools and managed questionnaires and tablets for fieldwork across Eastern and Northern Uganda.
- Assessed facility readiness for Health Information Exchange and paperless viral load requests, reviewing UgandaEMR records against ART registers.
- Conducted key informant interviews and FGDs with clinical and data staff and checked data quality across HMIS tools and the EMR.
- Documented constraints affecting EMR use, including DHIS2 knowledge, internet and power stability, and computer performance, and shared facility-level feedback at each visit.
- Compiled progress and activity reports with safe backup and storage, completing a 12-day assessment in September 2023 within a seven-region exercise covering Kigezi, Ankole, Lango, Acholi, Busoga, Bugisu, and Karamoja.

Research Associate, September 2021 - January 2023

Bronkar (U) Limited, Kampala, Uganda

- Developed assignment workplans and tracked deliverables across research assignments.
- Designed methods and survey tools for qualitative and quantitative collection.
- Trained enumerators and conducted field research including interviews and structured surveys.
- Cleaned and analyzed survey data and prepared findings for client reports and proposals.

Research Assistant, April 2021 - June 2021

Makerere University Agricultural Research Institute, Wakiso, Uganda

- Collected data from field trials and cleaned data in Excel for analysis.

Data Entry Clerk, January 2021 - February 2021

UNYFA, Kampala, Uganda

- Entered and checked survey data for organizational databases and ran quality audits to keep records complete before analysis.

PORTFOLIO

- GitHub and portfolio: github.com/mugarrajohnson and johnsonmugarra.com, with repositories, project write-ups, live D3.js analytics, and publications with DOIs.
- Vanilla forecasting: Prophet and ARIMA models combining NASA POWER climate data with field measurements; interactive harvest model on the website, presented nationally at Makerere.
- Farm data platform (Enimiro): seven scheduled Python and R pipelines moving SurveyCTO submissions into PostgreSQL, feeding Power BI and Looker dashboards; available as a private walk-through on request.
- Automated evaluation pipeline: Python ingestion, cleaning, and report generation that makes the recurring evaluation cycle repeatable.

EDUCATION

M.Sc. Data Science, September 2022 - October 2024

Manipal Academy of Higher Education (MAHE), Karnataka, India

Coursework: Machine Learning, Deep Learning, Predictive Analytics, Time-Series Forecasting

B.Sc. Agriculture Science, August 2016 - May 2021

Makerere University, Kampala, Uganda

CERTIFICATIONS

- SAS Certified Statistical Business Analyst
- Microsoft Certified: Power BI Data Analyst Associate
- Tableau Desktop Specialist
- Google Project Management Certificate
- Human Subjects Research (CITI)

REFERENCE

- Available on request.